Iterative Method of Isolating Primary and Secondary Components of Remanent Magnetization Illustrated by Using the Upper Devonian Catombal Group of Australia

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Directions of natural remanent magnetization from Upper Devonian Catombal Group red beds retain after magnetic cleaning a large secondary component acquired after the rocks were folded. A computer program was developed to eliminate progressively this 'hard' secondary component, which was found to lie along the present dipole field direction with an average intensity 1.6 times that of the primary component. It is interpreted to be due to weathering. The best estimate of primary direction was taken as the one in which the site mean directions, after removal of some secondary component and unfolding, gave minimum dispersion. The Upper Devonian field direction at Wellington derived from this procedure is (171°, −14°), with A_{95} = 8°. The southern hemisphere paleomagnetic pole (49°S, 312°E) obtained is not significantly different from the only published Australian pole position of Upper Devonian age, from the Lochiel Formation, at (58°S, 320°E).

INTRODUCTION

Well-documented and well-tested remanent magnetization directions from Carboniferous and more recent Australian rocks [*Hicken et al.*, 1972; *McElhinny and Embleton*, 1974] yield a useful polar path relative to eastern Australia, but with increasing geological age the path becomes more and more nebulous. A reliable apparent pole position from the Catombal Group would extend the more precise part of the polar path back to the Upper Devonian.

A fold test [*Graham*, 1949], using sites with different bedding tilt (see Figure 1), showed that directions relative to bedding (primary) (Figure 2*a*, circles) were more scattered than the same directions plotted by using the present horizontal (secondary) as the reference frame (Figure 2*a*, squares). This test showed that a large part of the remanent magnetization had been acquired after the folding. Alternating field demagnetization indicated that both primary and secondary magnetization had a high coercivity.

This paper describes a method of stripping off the secondary component analytically. Minimum scatter of site magnetization directions relative to the bedding plane at the site is used as the criterion for determining the amount of secondary component to be subtracted. A range of possible secondary directions was tried, from that of the Carboniferous direction (soon after folding) to that of the present dipole field.

The resulting great circles showing the rotation of mean site directions are illustrated in Figure 3. It will be seen that small differences in secondary direction yield quite large changes in minimum dispersion.

GEOLOGY

The Upper Devonian Catombal Group of central western New South Wales, Australia, forms a meridional belt 150 km long from Cargo to Wellington. The red sediments sampled occur in the Molong-Wellington area and outcrop over a strike length of 75 km with an average width of 7 km. These sediments are red sandstones, shales, and conglomerates and form a series of rugged hills called the Catombal Range. The sediments of the Catombal Range overlie, in places conformably, limestones and marls of Lower to Middle Devonian age.

The structure of the Catombal range between Wellington and Molong is characterized by tight north south trending synclines and anticlines. The dominantly synclinal overall structure is preserved in the central region. The region near Molong, however, is characterized by thin, elongated, well-preserved synclines and anticlines. The folding has resulted in wide variations in dips of beds in different areas, yielding ideal conditions for a fold test suitable for evaluating the timing of the acquisition of the remanent magnetization. The folding is in places nearly isoclinal, and most of the sites are on parallel sides of steep folds, a situation which reduces the possibility of significant errors in unfolding. The axes of the folds have low plunges, so that unfolding about the strike at the sites will only cause a small error. Unfolding about the axis of the folds [e.g., *Cruden and Charlesworth*, 1972] would remove this error and increase the precision of the unfolded data, but the necessary field information is not available; in fact, it was commonly a major problem to define the top of the beds, in places overturned at the sites. The locations of the 13 collection sites are shown in Figure 1. Nine collection sites were in the lacustrine Curra Creek Conglomerate, three in the marine Brymedura Sandstone, and one in the fluvial Kurrool Formation. Stratigraphic names are from *Conolly* [1963].

METHOD

The primary direction of magnetization is sought by an iterative computer simulation which involves at least two steps. The first obtains an approximate primary direction, and the second uses this approximate direction to produce a better-defined direction. This direction can then be used in a third estimate, if necessary, and so on.

The initial approximate primary direction is obtained by systematically reducing the intensity of magnetization along the assumed secondary direction and applying a fold test [*Graham*, 1949] to the resulting directions after each stage of the reduction. If a significantly improved grouping of site means after unfolding is obtained by reducing the component along an assumed direction, then the approximate primary direction

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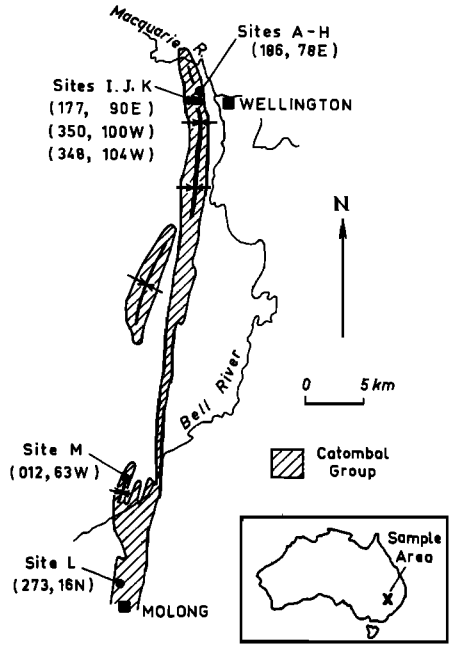


Fig. 1. A generalized geological sketch map of the northern part of the Catombal Group after *Conolly* [1963] showing the position of the paleomagnetic collection sites and, in parentheses, the strike and dip of the beds at the sites.

thus obtained is used to calculate the proportion of secondary component at each site, and this new value of secondary component is used in a second set of calculations.

The in situ magnetic vector $\mathbf{R}_t$ for each site can be expressed as

$$\mathbf{R}_t = \mathbf{P}_{B_t} + \mathbf{S}_t \qquad (1)$$

where $\mathbf{P}_{B_t}$ is the in situ primary vector at each site which has been rotated by the deformation of the rocks at the site and $\mathbf{S}_t$ is the secondary vector at each site. Each reduction of intensity along a specified direction $\mathbf{S}$ can be represented by

$$\mathbf{R}_t' = \mathbf{P}_{B_t} + \mathbf{S}_t - k\hat{\mathbf{S}} \qquad (2)$$

where k is a scalar multiplicative factor and $\hat{\mathbf{S}}$ is the unit vector in the direction of $\mathbf{S}$. From this it can be seen that as $k \to \infty$, $\mathbf{R}_t' \to -k\hat{\mathbf{S}}$ (Figure 2*b*), i.e., the resultant direction approaches the opposite direction to the one assumed. The plot of $\mathbf{R}_t'$ on a stereographic net (Figure 2*b*) will thus be a segment of a great circle from $\mathbf{R}_t$ through $\mathbf{P}_{B_t}$ to $-\mathbf{S}$. When a bedding correction is applied, the path representing the reduction of intensity along $\mathbf{S}$ will, of course, also be a segment of a great circle. In this case, however, with progressive subtraction of intensity along $\mathbf{S}$, the resultant direction of remanent magnetization for each site will approach a limiting position $-\mathbf{S}_{BC_t}$. This corresponds to the direction obtained by applying the bedding correction at that site to $-\mathbf{S}$, the direction in the reverse direction to the specified $\mathbf{S}$ direction.

An initial estimate ($\hat{P}_1$) of the $\mathbf{P}$ direction is obtained as follows. Possible $\mathbf{S}$ directions are obtained by reference to the apparent polar wander curve (and the reverse of the apparent polar wander curve) since the time of deformation of the rocks. The magnetic intensity is then reduced along this direction by an increment proportional to the average magnetic intensity of the formation. A bedding correction is then applied, and the dispersion of the resultant site directions of magnetization is measured. This procedure is repeated until the resultant direction of magnetization at each site approaches its limiting position $-\mathbf{S}_{BC_t}$, which is explained above. This is repeated for each possible $\mathbf{S}$ direction. If a well-defined minimum dispersion is obtained with the use of one of the possible $\mathbf{S}$ directions (as in this case, when the present dipole field direction is used), it is assumed that this represents the true $\mathbf{S}$ direction, and the average of the resultant directions corresponding to the minimum dispersion after bedding cor-

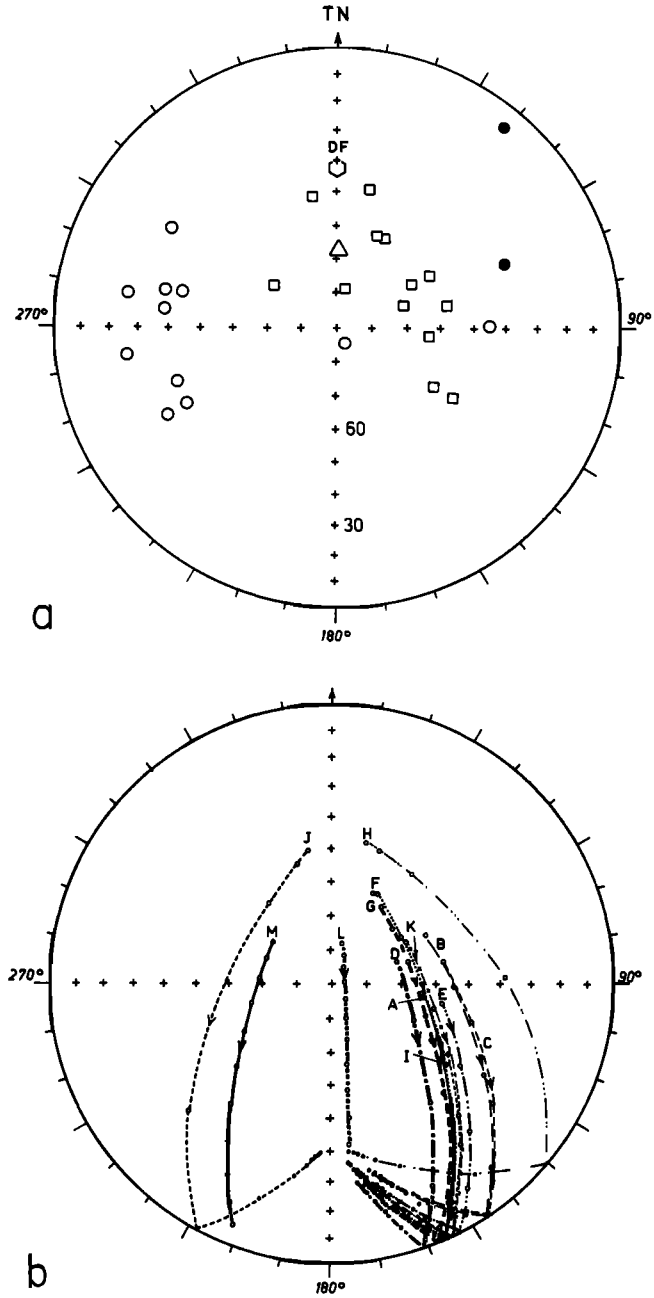


Fig. 2. (*a*) An equal area net showing the site mean directions for the Catombal Group before (squares) and after (circles) correction for the dip of the bedding. The better grouping relative to the horizontal (squares) indicates a large postfolding component. The group (circles) on the left of the net is from a group of sites with similar dips and strikes. *Green*'s [1961] mean Catombal Group direction with respect to the horizontal is also given (triangle). Open circles represent positions in the upper hemisphere, closed circles positions in the lower hemisphere. (*b*) An equal area net showing the trace of the paths of the mean site directions, with respect to the horizontal, when 25% increments of mean intensity are successively removed from the dipole field direction (0, −42) at Wellington.

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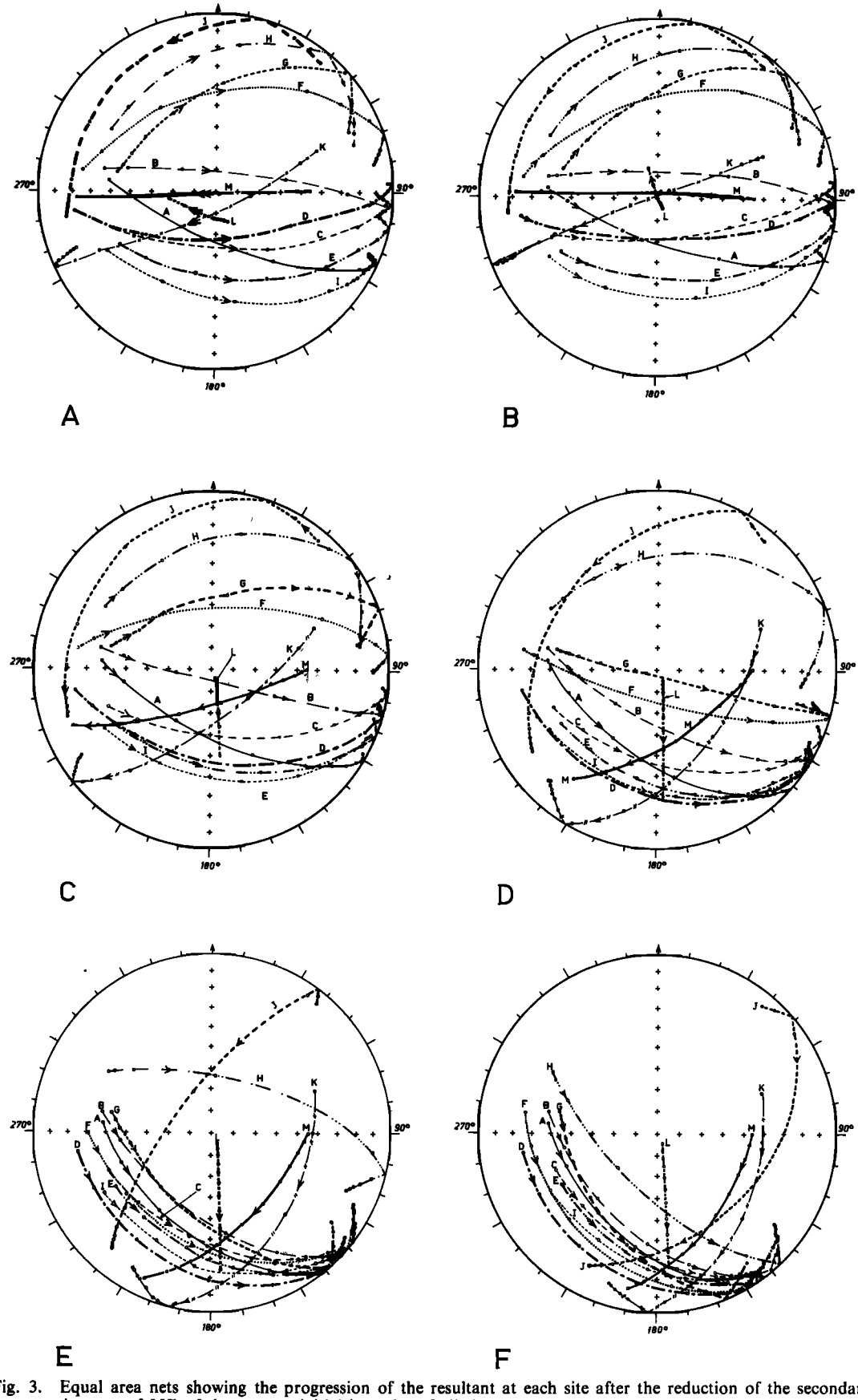


Fig. 3. Equal area nets showing the progression of the resultant at each site after the reduction of the secondary component in steps of 25% of the average initial intensity of all sites. Conventions for circles are as for Figure 1*a*. All directions are relative to bedding. The directions of the vector removed are as follows: (*a*) (023, −78), Upper Carboniferous (see Table 4); (*b*) (024, −80), Jurassic (see Table 4); (*c*) (008, −72); (*d*) (005, −62); (*e*) (002, −52); and (*f*) (000, −42), dipole field direction at Catombal. Directions for (*c*), (*d*), and (*e*) are approximately $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{3}{4}$ of the way along the great circle from the pole position during the Jurassic to the dipole position today, respectively.

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TABLE 1. Intensities in **P** and **S** and the Error Vector |**E**| Obtained by Resolving **R** for Each Site

Site	**P**	**S**	|**E**|	*P/S*	*E/S*
A	61	72	3	0.85	0.04
B	30	44	5	0.68	0.11
C	28	35	4	0.80	0.10
D	37	52	5	0.71	0.10
E	42	46	1	0.91	0.02
F	16	36	1	0.44	0.03
G	31	73	3	0.42	0.04
H	5	34	4	0.15	0.12
I	40	40	3	1.00	0.08
J	5	33	3	0.15	0.09
K	18	31	9	0.58	0.32
L	79	128	23	0.61	0.18
M	45	71	19	0.63	0.27
A–M				0.61	0.12

rection is used as the initial estimate ($\hat{P}_1$) of the primary direction of remanent magnetization. Where large variations occur between the average intensities at different sites, the statistical point of minimum dispersion may not be an adequate criterion for the initial estimate ($\hat{P}_1$) of the **P** direction; it may then be preferable to make the initial estimate by simple inspection of the plotted reduction curves (such as those of Figure 3).

With the use of the assumed secondary direction **S** and the initial estimate ($\hat{P}_1$) of the primary direction the component of the site magnetization vectors in the **S** direction can be calculated. The in situ magnetization vector at each site can be expressed as

$$\mathbf{R}_i = \mathbf{P}_{1B_i} + \mathbf{S}_i + \mathbf{E}_i \qquad (3)$$

where $\mathbf{P}_{1B_i}$ is the component in the direction defined by applying the reverse of the bedding correction at each site to the $\hat{P}_1$ direction (to account for the structural deformation), $\mathbf{S}_i$ is the component in the assumed **S** direction, and $\mathbf{E}_i$ is the component in a direction perpendicular to the plane containing $\mathbf{P}_{1B_i}$ and $\mathbf{S}_i$ (see Table 1). The $\mathbf{E}_i$ term represents that component which cannot be resolved into the $\mathbf{P}_{1B_i}$ and $\mathbf{S}_i$ directions. It can be thought of as an error term resulting from inaccuracies in the $\hat{P}_1$ and **S** vectors as well as those entailed in the orientation of samples at each site.

A second estimate ($\hat{P}_2$) of the primary direction of magnetization is now made. This is done by progressively reducing the intensity of magnetization at each site along **S** by increments proportional to $\mathbf{S}_i$, applying a bedding correction at each site, and testing for minimum dispersion. This procedure is similar to that employed in obtaining $\hat{P}_1$. In this case, the great circular paths traced can be described by

$$\mathbf{R}_{BC_i} = \mathbf{P}_{2B_i} + \mathbf{S}_{BC_i} + \mathbf{E}_{BC_i} - K|\mathbf{S}_i|\ \hat{\mathbf{S}}_{BC_i} \qquad (4)$$

where $\mathbf{R}_{BC_i}$ is the resultant direction after each reduction corrected for bedding, $\mathbf{S}_{BC_i}$ is the calculated component in the **S** direction at each site corrected for bedding, $\mathbf{E}_{BC_i}$ is the component in the $\mathbf{E}_i$ direction for each site corrected for bedding, K is a multiplicative factor incremented for each reduction, $|\mathbf{S}_i|$ is the absolute value of the calculated intensity in the **S** direction at each site, and $\hat{\mathbf{S}}_{BC_i}$ is the unit vector in the direction obtained by correcting for bedding the **S** direction at each site. As has been the case previously, when a well-defined minimum dispersion is obtained, the average of the resultant directions corresponding to the minimum dispersion is taken to be the estimate of the primary direction of magnetization ($\hat{P}_2$).

The above procedure can be repeated in order to obtain successive and more accurate estimates of the primary direction of magnetization. In the present case the location and precision of the calculated primary directions changed little after two iterations.

The subjectivity of the selection of the direction of the secondary component can be overcome by trial and error of all possible secondary components between the age of the formation and the present, obtainable from the polar wandering curve of the appropriate block. As the true secondary direction is approached, the minimum dispersion after reduction along **S** and correction for bedding will reach a minimum; departures from the true secondary direction will show as increased scatter at minimum dispersion. Table 2 shows the mean directions and dispersions for the Catombal sites using in turn the Carboniferous, Jurassic, present dipole field, and three intermediate directions for the secondary direction. It may be seen that quite small differences in choice of secondary direction yield quite large differences in the resulting primary direction and scatter. Therefore the true secondary direction can be accurately selected by taking sufficient possibilities.

TABLE 2. Directions at Optimum Precision [*Fisher*, 1953] After Vector Reduction for Different Directions of Secondary Component

Secondary Component Reduced	Percent of R_1 Taken Off	*D*, deg	*I*, deg	*k*	α_{95}, deg
		Carboniferous			
023, −78	25	278	−60	2.4	35
		Jurassic			
024, −80	50	297	−77	2.4	35
008, −72	75	130	−67	2.4	35
005, −62	150	135	−19	3.1	28
002, −52	150	150	−15	5.2	20
		Dipole			
000, −42	150	159	−13	9.3	14
		After iteration			
000, −42	110	160	−10	24.0	8.6

R_1 is the magnitude of the mean intensity of the vector after cleaning in 100 Oe. *D* is the mean group declination east of true north; *I* is the inclination negative upward; *k* is the best estimate of *Fisher*'s [1953] precision; and α_{95} is the half angle of the cone of confidence at a probability $P = 0.95$.

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TABLE 3. Site Directions Relative to the Horizontal

Site	N	Natural Remanent Magnetization					100 Oe				
		D, deg	I, deg	k	α, deg	⟨Int⟩	D, deg	I, deg	k	α, deg	⟨Int⟩
A	5	104	−65	6.4	51	25.9	96	−63	5.9	35	28.3
B	5	70	−59	45.8	11	28.6	61	−59	88.1	8	28.9
C	5	86	−56	24.2	16	32.5	79	−57	25.6	15	28.9
D	6	70	−75	6.25	29	28.7	71	−70	7.1	27	29.4
E	1	69	−56	...	...	28.9	102	−57	...	...	32.1
F	5	41	−51	42.5	12	25.1	27	−60	37.5	13	25.9
G	5	24	−69	11.9	23	25.0	25	−60	11.3	24	56.4
H	6	16	−47	111.1	6	33.1	14	−47	64.5	8	33.6
I	2	90	−53	74.2	8	19.2	121	−50	21.8	56	20.9
J	6	340	−49	52.6	9	27.3	350	−50	120.0	6	30.1
K	10	66	−68	5.0	24	25.7	61	−65	4.3	27	24.9
L	4	0	−77	86.1	10	167.4	14	−78	425.1	5	114.6
M	6	318	−69	5.3	32	127.9	305	−69	6.2	29	79.1
A–M	13	47	−67	13.9	12	...	51	−68	12.1	12	...

The site letters refer to the positions of Figure 2; N is the number of separately oriented samples; D is the declination east of true north; I is the inclination positive downward; k is *Fisher*'s [1953] best estimate of precision; α is the half angle of the cone of confidence at $P = 0.05$; and ⟨*Int*⟩ is the arithmetic mean intensity in emu/cm^3 × 10^{-6}.

Two advantages of this method of removing secondary components, compared with standard thermal and magnetic cleaning, deserve mention. First, it is possible, by using our iterative method, to remove a secondary component with a higher coercivity (impossible with magnetic cleaning) or a higher blocking temperature (impossible with thermal cleaning) than that of the primary component with which it is physically associated, as long as the secondary direction is known. This procedure is advantageous in partly weathered rocks, commonly oxidized, in which the weathered material is likely to be hematite, with a higher coercivity and blocking temperature than the primary component (if magnetite).

Second, if a secondary component is suspected, different possible directions for it, related to the time of formation as deduced from geological reasoning, may be tested against each other for minimum dispersion. The limitation of the method is that it can only be used on a group of sites in which the bedding is tilted at different known angles. Even in a suitable tectonic environment it is no simple matter to find a selection of suitably dipping sites.

RESULTS

The clustering of site means (Figure 2*a*), both before and after magnetic cleaning, relative to the present horizontal at each site (Table 3) indicates that a large proportion of a high-coercivity component was acquired after the folding. Demagnetization of at least one specimen from each site up to 600-Oe peak field indicated that direction changes in cleaning up to 600 Oe were also negligible. It is thus evident that a 'hard' secondary component has been introduced at some time after folding.

The most likely times for introduction of a secondary component into the Catombal Group are soon after folding, at a time of maximum stress and possibly raised temperature, during a phase of laterization, or over an extended period up to the present, when weathering may be expected. The first of the above possibilities would result in an Upper Carboniferous age for **S**, and the second in a Jurassic age for **S**. The third age for **S** is along the dipole field direction.

An average direction and magnetization for the Upper Carboniferous of eastern Australia were calculated as $D = 023°$, $I = -78°$ by using the poles given in Table 4. The method described in the previous section for obtaining an estimate of the secondary direction (**S**) of magnetization was applied to the site average magnetization vectors with the use of this direction. The maximum decrease in dispersion which resulted after unfolding was less than 10%. However, a grouping into two almost opposite directions occurs when excessive amounts of the **S** components are removed. This does not reflect a field reversal but simply the common occurrence at the 13 sites of

TABLE 4*a*. Magnetization Directions

Rock Unit	N	D, deg*	I, deg*	k	α_{95}, deg	Reference	Catalog Number
Upper Carboniferous							
Percy Creek Volcanics	1	064	−85	...	...	*Chamalaun* [1968]	6161
Paterson Toscanite	4	002	−67	...	...	*Irving* [1966]	659
Upper Kuttung Sediments	18	011	−80	45	5	*Irving* [1966]	676
Mean Upper Carboniferous	23	023	−78	...	...	Sites units	
Jurassic							
Noosa Heads Intrusion	4	051	−70	48	13	*Robertson* [1963]	7028
Gibraltar, Prospect, Ginginbullen	3	009	−82	...	...	*Boesen et al.* [1961]	5020, −21, −19
Tasmanian Dolerite	51	319	−84	111	3	*Irving* [1963]	7026
Mean Jurassic	3	024	−80	...	...	Groups units	

N is the number of sites; D is the declination east of true north; I is the inclination positive downward; k is *Fisher*'s [1953] best estimate of precision; and α_{95} is the half angle of the cone of confidence at a probability $P = 0.95$. The catalog numbers are taken from *Hicken et al.* [1972].

*The declinations and inclinations have been inverted, where it was necessary, so that they are in a sense to yield a reversed magnetic field.

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TABLE 4b. Pole Positions

Rock Unit	Age	South Pole	A_{95}, deg	Reference	Catalog Number
Mugga Mugga Porphyry	Upper Silurian (423)	80°S, 340°E	7	*Briden* [1966]	4.2
Mereenie Sandstone	Silurian(?)-Devonian	41°S, 040°E	10	*Embleton* [1972]	5.1
Bowning Group	Lower Devonian	64°S, 045°E	9	*Luck* [1973]	5.2
Ainslie Volcanics	Lower Devonian	71°S, 353°E	10	*Luck* [1973]	5.3
Lochiel Formation	Upper Devonian	58°S, 320°E	7	*Luck* [1973]	5.5
Catombal with respect to horizontal	Upper Devonian	49°S, 103°E	19	This paper	
Catombal with respect to beds*	Upper Devonian	49°S, 312°E	8	This paper	

A_{95} is the half angle of the cone of confidence at a probability $P = 0.95$ using site poles as units. The catalog numbers are taken from *McElhinny and Embleton* [1974].

*The declinations and inclinations have been inverted, where it was necessary, so that they are in a sense to yield a reversed magnetic field.

near-vertical tilts in opposite directions about a common axis (Figure 1). As the introduced reverse **S** vector (with respect to horizontal) is increased, the site directions approach it asymptotically, and the bimodal grouping (with respect to bedding) is caused by the bedding corrections of similar size and opposite sense. Considering Figure 3*a* we conclude that the Upper Carboniferous direction did not represent the secondary direction of magnetization.

The second likely time for the emplacement of a secondary component is during a Jurassic laterization. An average Jurassic direction of $D = 024°$, $I = -80°$ was calculated by using the poles given in Table 4. The results of removing **S** in this direction gave dispersed results similar to the above (Figure 3*b*).

Three additional possible secondary directions on the great circle at intervals of approximately 10° between the above direction and the present dipole field were also tried. These gave results which grouped better as the present field direction was approached (Figure 3*c*, 3*d*, and 3*e*). The third likely geological time mentioned above for the emplacement of a secondary component is Recent, along the dipole field, which, at the latitude of the sites near Wellington, is $D = 0°$, $I = -42°$. When this direction was used in an initial estimate of the primary direction (**P**) of the average site magnetization vectors, the α_{95} [*Fisher*, 1953] reached a minimum less than half the original after reduction of the secondary direction components by 160% of the intensity of **R** (Figure 3*f*).

For comparison, the same reduction was performed without applying a bedding correction. In this case the dispersion of the sites increased steadily to define a maximum dispersion and then decreased as the directions moved asymptotically toward the position obverse to the present dipole field direction specified (see Figure 2*b*). Thus the significant decrease in dispersion in the previous case was clearly related to the bedding correction and not to the inclusion of a dominating component of constant direction relative to the horizontal frame of reference at each site.

The estimate of the direction of **P** obtained by assuming **S** in the direction of the present field, was (159°, −12°, $\alpha_{95} = 14$). This direction was used in conjunction with the assumed **S** and the in situ magnetization vector at each site to obtain better estimates for each site of the intensities of the **S** component. These new site **S** values were then used instead of the formation average intensity to obtain a better estimate of the primary direction. In this case a discrete minimum was obtained at 110% reduction of the new site **S** values, giving a formation direction of (160°, −10°, $\alpha_{95} = 8.6$), not significantly different in direction, but more precisely defined. A second estimate of secondary intensities was made with the use of this direction. The resulting primary direction obtained was (161°, −11°), with $\alpha_{95} = 8.4$, which is essentially the same as the previous estimate.

Of the 13 Catombal Group sites, eight have to be unfolded by easterly (anticlockwise) rotations, four by westerly (clockwise) rotations, and one by a rotation to the south. Sites A–H all dip in a similar direction and are likely to cause a bias in the mean from an error common to all in the unfolding process (see Figure 3). In an attempt to overcome this bias, sites A–H, which are closely spaced, have been reduced to three supersites, which have been combined with the other five sites. A new set of iterations (Table 5) has been performed on these eight sites. The mean direction and pole quoted are after the third iteration, but there is no significant difference between the first, the second, and the third iterations.

The intensities of magnetization along the assumed primary and secondary directions at each site and along a direction perpendicular to the primary and secondary directions are given in Table 1. These indicate fairly uniform primary and secondary intensities of magnetization, with the primary magnetization being smaller than the secondary in all but one of the 13 sites and on average having an intensity about 0.6 of the secondary intensity. This consistency in the relative strengths of the primary and secondary magnetizations for most sites would also tend to support the assumption of the dipole field direction as that of the secondary component.

Also, the small size of the error vector $|\mathbf{E}|$ (Table 1) at most sites indicates that our **P** and **S** account quite well for the initial data. Larger $|\mathbf{E}|$ at sites L and M may be associated with larger unfolding errors at sites with lower bedding dips.

DISCUSSION

The apparent pole position of the Catombal Group, using our iteration method, is 49°S, 312°E ($A_{95} = 8$). It is considered to give a good estimate of the position of the South Pole relative to eastern Australia in Upper Devonian time.

It is encouraging to note with respect to the present study

TABLE 5. Iterations Based on Eight Sites and a Secondary Direction Along 0°, −42°

Iteration	D, deg	I, deg	k	α_{95}, deg
Initial	167	−26	9.5	19
1st	166	−18	28	11
2nd	169	−14	30	10
3rd	171	−14	33	10

The pole position for the third iteration was 312°E, 49°S, and α_{95} was 8°.

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that in testing the various geologically reasonable directions of secondary magnetization a secondary direction among these was found which gave a uniquely defined minimum dispersion in the resultant primary direction. This fact implies that with suitable geological constraints on the secondary direction this method may be able to produce a unique solution for the primary direction. In the present study, only primary components of the same polarity were present. It is possible, however, for primary components in normal and reversed directions to be present, and because of this it is imperative to inspect visually the Phase 1 reduction plots, since such a case may well result in worse dispersion overall. This serves to indicate that the present method is not automatic and could be abused if its interpretive content is ignored.

Pole positions from all previously published A group (more reliable) data from *McElhinny and Embleton* [1974] of Devonian formations of eastern Australia, and their associated error ovals at $P = 0.95$, are shown in Figure 4. The details of location and circles of confidence are given in Table 4. It should be noted that the time span involved is considerable (about 100 m.y.), and two of the formations may be Upper Silurian.

Only two of the four previously calculated pole positions are consistent with each other, and only the Lochiel Formation [*Luck*, 1973] is consistent with that obtained in the present study. This inconsistency may either be due to the differences in age or be caused by secondary components that form part of the remanent vector for the formation studied and have remained undetected because no field test was available.

When an average drift rate of 3 cm/yr, consistent with mean spreading rates of present day ocean ridges, is assumed, a rate of drift of 25° per 100 m.y. results. If we allow a possible error of 50 m.y. in the quoted ages of the formations used, we can permit drift movement of 30° between the Bowning and Catombal group poles. If the movement was from the Bowning Group pole (64°S, 045°E) to the Catombal Group pole (49°S, 312°E), all the pole positions are consistent with each other except the Mereenie Sandstone pole, and since it was derived from rocks in Central Australia, it may have been separated from the others at the time by a rift or trench.

Alternatively, it may be noted that the previously estimated pole positions have latitudes between that of the pole calculated for the Catombal Group in this study and that of the present earth's field. This fact suggests that a component in the direction of the present earth's field may be incorporated to some degree in the results shown in Figure 4. For comparison, the calculated pole position before the unfolding of the beds is also shown for the Catombal Group. Because of the presence of primary components dispersed by large differences of tilt at the sites the confidence oval for this pole is considerably enlarged. This pole with its error oval gives some indication of the large errors possible, when field criteria of stability are not available, where significant secondary components are present.

An even more serious case may occur where no field criteria

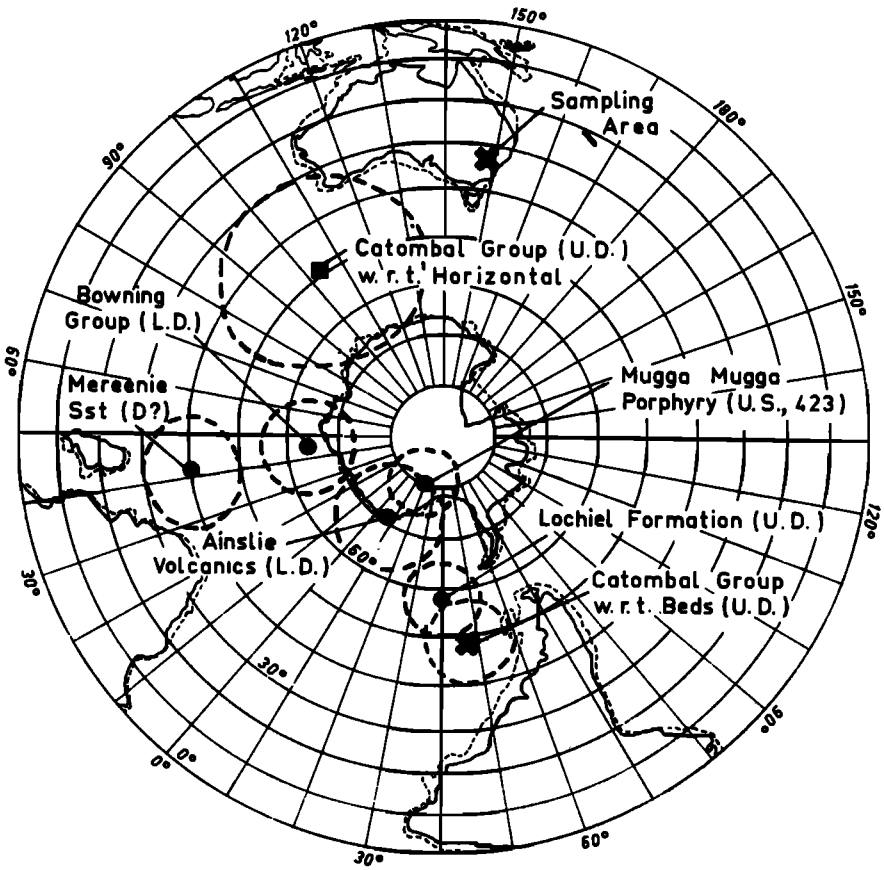


Fig. 4. An equal area polar projection of the southern hemisphere of the earth showing published pole positions with their $A_{95}°$ ovals of confidence for Australia for 'Devonian' time (see Table 4). Also shown are pole positions derived from the Catombal Group in this paper relative to the horizontal and relative to the bedding. The latter is the pole derived by the vector reduction technique after the removal of 110% of R_1 in the direction of the dipole field (000, −42), using eight sites and three iterations.

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are available to rule out the possibility of large secondary components. In this case, tightly defined resultant directions from horizontal beds could be confused with primary directions. If the secondary component has a higher coercivity spectrum or blocking temperature spectrum than the primary component, the latter would be removed preferentially by laboratory cleaning. Where the respective spectra overlap, consistent but erroneous directions could also be obtained from laboratory measurements. This appears to be a possibility in areas of deep tropical and subtropical weathering, as in much of Australia, where 'hard' hematitic secondary components may be expected.

The analytical method presented here is not intended to replace the time-honored and normally adequate methods of laboratory demagnetization which are in use. It is thought, however, that the above approach may be helpful in resolving some 'hard' secondary from primary components in difficult geographical areas. Further work is planned to compare the results of the above study with those obtained by using partial thermal demagnetization on samples from the same area.

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